

SABARINATH B

sabarinathb362@gmail.com · github.com/sabarinathb362 ·
linkedin.com/in/sabarinath-b-277a27197 · stackoverflow.com/users/18058330/sabarinathb362

Profile

Interested in AI, machine learning, computer vision, AI automation, and IoT, with hands-on experience developing AI-powered applications, intelligent systems, and computer vision solutions. Published IEEE research on AI-based wildlife monitoring, with a focus on practical, real-world applications.

Educational Qualifications

Amrita Vishwa Vidyapeetham, Kollam, Kerala — CGPA: 8.03 2024 – Present
Bachelor of Technology in Artificial Intelligence and Data Science

Sree Narayana Public School, Kollam — Score: 90% 2021 – 2023
Higher Senior Secondary – XII (CBSE)

Trinity Lyceum ICSE, Kollam — Score: 92.2% 2015 – 2021
Secondary School – X (ICSE)

Projects Undertaken

- **Fruit Freshness Detection using Self-Supervised Learning (2026) — PyTorch, SimCLR, ResNet-50**
Developed a self-supervised computer vision pipeline achieving **97.25% accuracy** with **3.7×** faster training efficiency for edge deployment.
- **IoT-Based Smart Energy Meter with Prepaid Billing, Theft Detection & Appliance Monitoring (2026) — ESP32, MQTT, Wokwi, Proteus**
Developed an ESP32-based smart metering system with real-time energy monitoring, appliance-level consumption tracking, prepaid billing, differential-current theft detection, and a live MQTT web dashboard.
- **GRU-Based Predictive Maintenance for Military Robotic Systems (2025) — Python, TensorFlow, GRU, SHAP**
Built an attention-based GRU model for robotic failure prediction, achieving **96.3% accuracy** and **100% recall** for failure detection.
- **Hybrid IoT-Based Wildlife Detection and Alert System (2025) — Python, YOLOv11s, NodeMCU**
Developed an infrared-triggered AI wildlife detection system with real-time IoT alerts, reducing unnecessary inference for efficient low-power deployment. *Published at IEEE SCEECs 2026.*

Internships

- **AI Research & Development Intern – CWEGE** Mar 2026 – Present
AMMACHI Labs, Kollam, Kerala
Working on **Large Language Models (LLMs)**, AI testing, intelligent automation, and AI-driven research focused on real-world applications and system evaluation.

Areas of Interest

- Artificial Intelligence and Machine Learning
- Computer Vision and AI Automation

Skills

- **Programming Languages:** Python, Java, JavaScript, R
- **Web Development:** HTML, CSS, React.js
- **Databases:** MySQL, MongoDB
- **Tools & Software:** Git, MATLAB, ANSYS Workbench, Blender, Adobe Photoshop, GIMP
- **Video Editing:** Adobe Premiere Pro, DaVinci Resolve
- **Game Development:** Unreal Engine, Godot
- **Soft Skills:** Structured Thinking, Creativity

Community Outreach Programs

- **Live-in-Labs® Volunteer** 2026 – Present
Chidambaranthapuram Village, Kerala
Community-based research and fieldwork for Chidambaranthapuram village.
- **Flood Relief Initiative** August 2026
Kuttanad, Kerala
Organized a flood relief initiative with friends, collecting and distributing essential supplies to families affected by the ongoing floods across various areas in Kuttanad, Kerala.

Certifications / Online Courses / Publications

- **A Hybrid Framework for Human–Wildlife Conflict Mitigation Using Infrared-Triggered YOLO and IoT Alerting**
IEEE SCEECS 2026, Published in IEEE Xplore
DOI: 10.1109/SCEECS65031.2026.11429959
- **Developing Large Language Models**
DataCamp (March 2026)
- **Deep Learning in Python**
DataCamp (February 2026)
- **MATLAB Onramp**
MathWorks (April 2025)
- **NPTEL – Data Science for Engineers**
Certificate ID: NPTEL25CS101S39700157

Workshops

- **Artificial Intelligence and Machine Learning Applications** – NIT Kurukshetra, August 2024
- **Artificial Intelligence and Machine Learning with Python** – NIT Calicut, September 2024
- **Data Mining** – NIT Calicut, September 2024
- **Hacktoberfest** – NIT Calicut, September 2024
- **amFOSS** – October 2024

Extra-curricular

- Table Tennis – Represented Kollam District in the Sub-Junior category at District-level tournament.

Hobbies

- 3D Modelling
- PC Gaming

Language Proficiency

- English – Read, Write, Speak, Understand
- Malayalam – Read, Write, Speak, Understand (Native)
- Hindi – Read, Write, Speak, Understand